Final Report - AI5200 Project 1 - Clay Chambers

1. Final models.

For each task, describe your final architecture and training recipe: depth, width, activation, normalization, optimizer, schedule, regularization, augmentation, and training time. A table works well here. Report test accuracy from the harness next to the baseline's.

MNIST

MNIST Final Model	
Depth	30
Width	128
Activation	SwiGLU
Normalization	LayerNorm at beginning of block
Optimizer	AdamW, lr=0.001, weight_decay=0.05
Schedule	CosineAnnealingLR, 15 epochs
Regularlization	standardization during augmentation
Augmentation	RandomResizedCrop, RandomRotation
Training Time	1.2 minutes
Baseline Accuracy	95.06%
Test Accuracy	98.94%

CIFAR10

CIFAR10 Final Model	
Depth	30
Width	128
Activation	SwiGLU
Normalization	LayerNorm at beginning of block
Optimizer	AdamW, lr=0.001, weight_decay=0.05
Schedule	CosineAnnealingLR, 30 epochs
Regularlization	standardization during augmentation
Augmentation	AutoAugment policy=CIFAR10, RandomHorizontalFlip,

CIFAR10 Final Model	
	v2.Normalization mean=0.5, std=0.25
Training Time	5.1 minutes
Baseline Accuracy	48.61%
Test Accuracy	58.83%

IMAGENET32

IMAGENET32 Final Model	
Depth	120
Width	128
Activation	SwiGLU
Normalization	LayerNorm at beginning of block
Optimizer	AdamW, lr=0.001, weight_decay=0.05
Schedule	CosineAnnealingLR, 30 epochs
Regularlization	standardization during augmentation
Augmentation	AutoAugment policy=IMAGENET, RandomHorizontalFlip, v2.Normalization mean=0.5, std=0.25
Training Time	42.2 minutes
Baseline Accuracy	6.77%
Test Accuracy	10.24%

2. What mattered most.

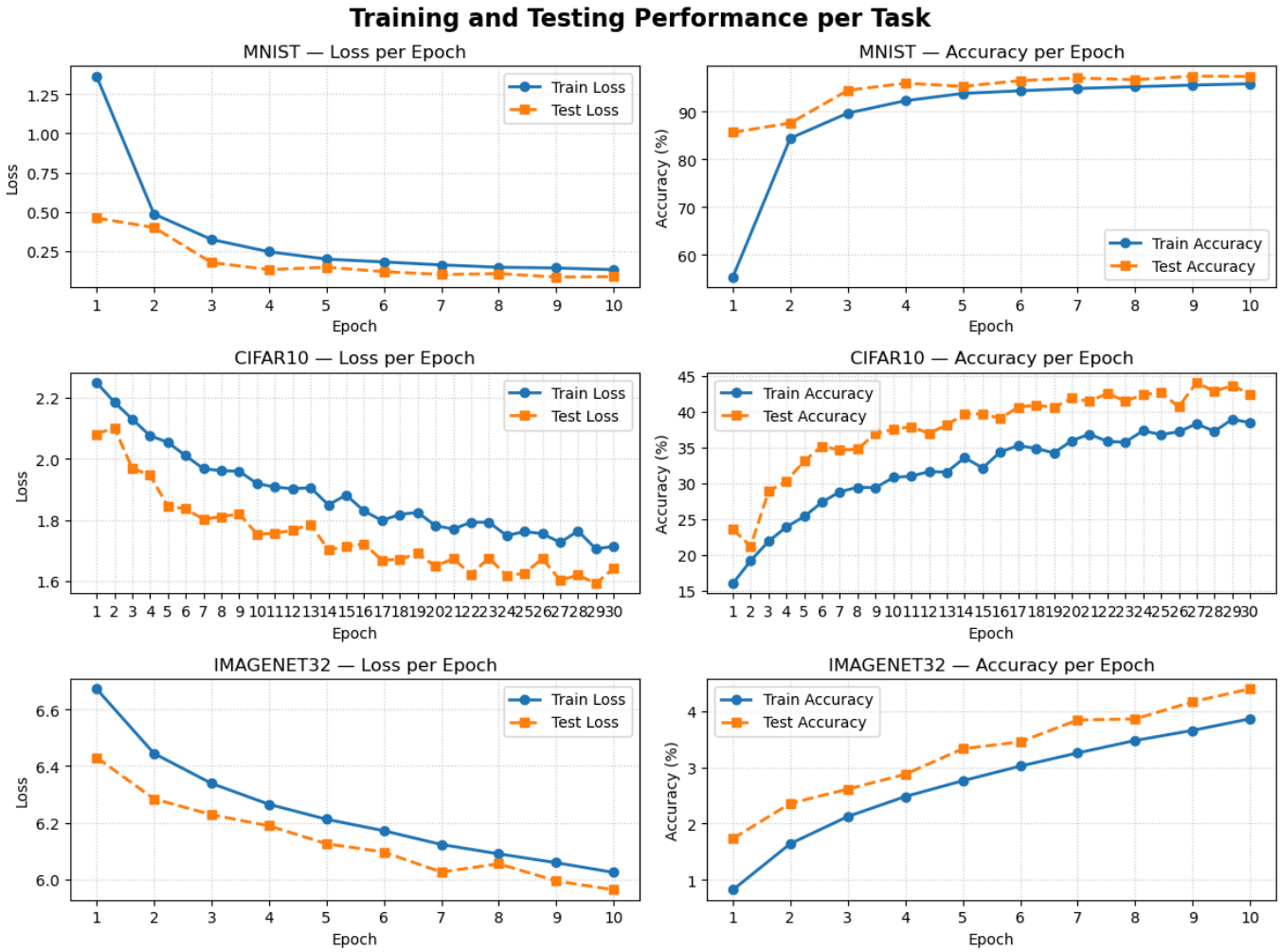
Which three to five changes improved accuracy the most? Support each claim with entries from your log that show the change and its effect.

1. Detecting and Correcting Overfitting with cifar10 - change 13

By far, the most important change I made was in Change 13 when I added accuracy and loss curves to my notebook. Up to then, I was almost exclusively watching the training loss and test accuracy but not understanding why accuracy was stagnant while loss dropped near zero. Once I added a visualization, it allowed me to have a eureka moment and finally start understanding that the networks were memorizing inputs rather than actually learning.


Training and Testing Performance per Task
MNIST — Loss per Epoch
Train Loss
Test Loss
Loss
Epoch
MNIST — Accuracy per Epoch
Train Accuracy
Test Accuracy
Accuracy (%)
Epoch
CIFAR10 — Loss per Epoch
Train Loss
Test Loss
Loss
Epoch
CIFAR10 — Accuracy per Epoch
Train Accuracy
Test Accuracy
Accuracy (%)
Epoch
IMAGENET32 — Loss per Epoch
Train Loss
Test Loss
Loss
Epoch
IMAGENET32 — Accuracy per Epoch
Train Accuracy
Test Accuracy
Accuracy (%)
Epoch

change 15



2. Data Augmentation - changes 14-16

Apart from addressing overfitting in general, data augmentation seemed to have the biggest impact on model accuracy. In Change 14, I began experimenting with different augmentation techniques. I realy wanted RandomAffine to work and spent several changes toying with it. Changes #14 - 16 were focued nearly entirely on augmentation.

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#### **CIFAR10 Mini Study**
When experimenting with Data Augmentation, I ran through many, many combinations
of pytorch v2 transforms with various parameters. I spent a little time building a
demo notebook so I could visualize what the transforms were actually doing. It's
in my repo. Here's what I learned.

First, I saw published means and standard deviations for imagenet32 on the
[documentation page for pytorch]
(https://docs.pytorch.org/vision/0.29/transforms.html). So, I looked up published
values for cifar10 and found them. However, those values are derived from the
whole dataset. In a real-world scenario, we wouldn't know the inference values in
advance, so I'm not planning to use them in my real model beyond this initial mini
study.

**Published means and std devs**
    cifar10_mean = [0.4914, 0.4822, 0.4465]
    cifar10_std  = [0.2470, 0.2435, 0.2616]
```

```
| Normalizations | Loss<br>after 10 epochs | Train Accuracy<br>after 10 epochs |
Test Accuracy<br>after 10 epochs |
| :--- | :--- | :--- | :--- |
| None (pixel values 0.0 - 1.0)  | 1.36 | 51.06 | 48.50 |
| v2.Normalize m = 0.5, std = 0.25         | 0.39 | 86.38 | 50.21 |
| v2.Normalize m = 0.5, std = 0.50         | 0.49 | 82.73 | 49.51 |
| v2.Normalize m = published, std = published   | 0.35 | 87.43 | 49.61 |


**Combine v2.Normalize with other augmentations**
| Normalized <br>plus other<br>transforms | Loss<br>after 10 epochs | Train
Accuracy<br>after 10 epochs | Test Accuracy<br>after 10 epochs |
| :--- | :--- | :--- | :--- |
| v2.AutoAugment  | 1.51 | 46.76 | 48.93 |
| v2.RandomHorizontalFlip         | 0.75 | 73.74 | 51.65 |
| v2.RandomAffine         | 1.46 | 47.48 | 46.56 |
| v2.RandomResizeCrop   | 1.62 | 41.82 | 48.53 |
| v2.RandomResizeCrop<br>and RandomHorizontalFlip   | 1.64 | 40.71 | 47.96 |
| v2.AutoAugment<br>and RandomHorizontalFlip   | 1.52 | 43.58 | 51.18 |


*_Label: CIFAR-10 Mini Study_*

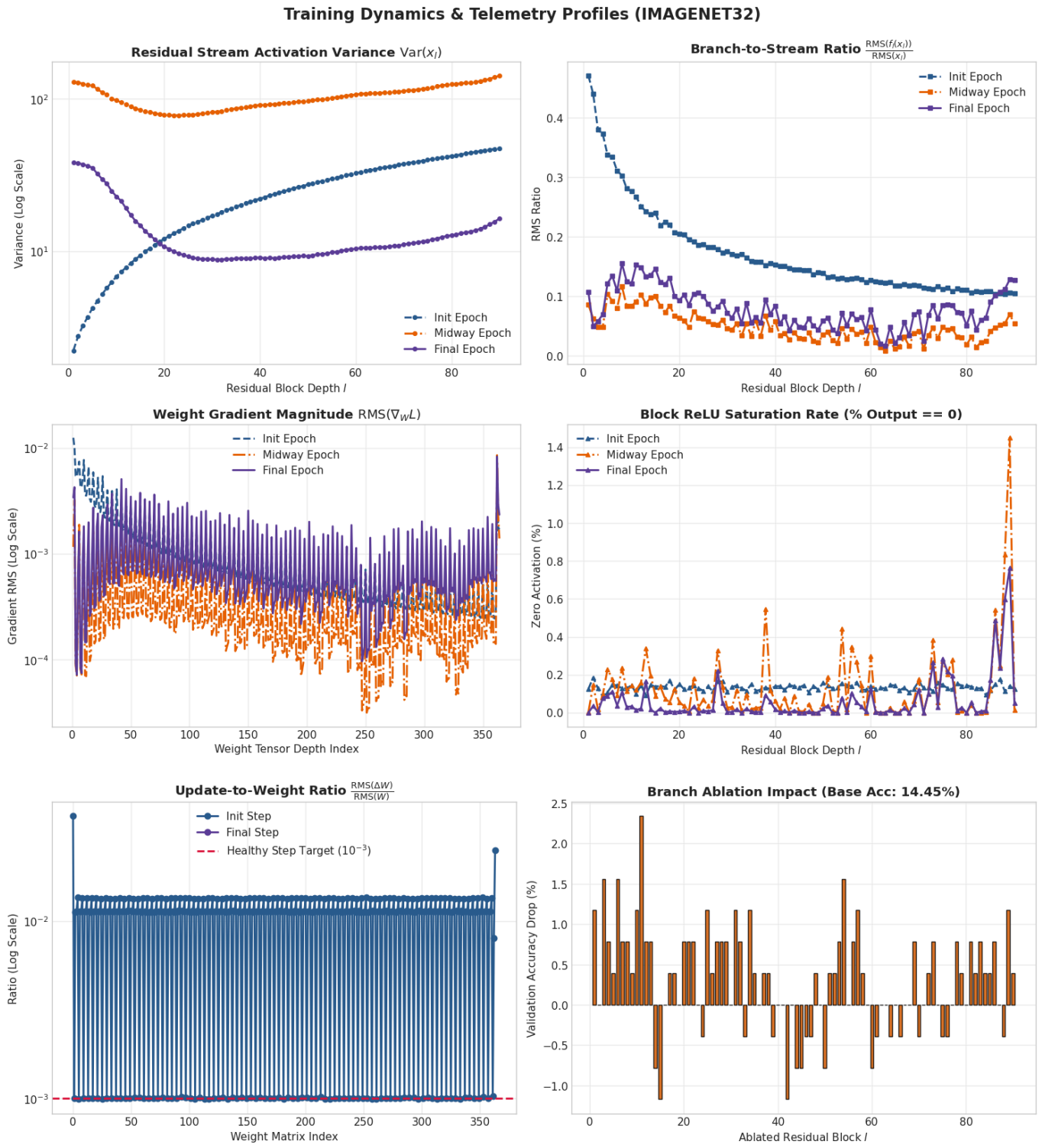
*Conclusion of mini study* It appears that the values used in normalization do
matter. And it appears that HorizontalFlip alone isn't enough to mitigate
overfitting. But it's useful as part of a larger data augmentation strategy. It's
not clear to me yet which strategy will yield the best results. Going forward, I'm
going to combine RandomHorizontalFlip and AutoAugment as a starting recipe on top
of normalization with mean = 0.5, std = 0.25. It doesn't seem to learn quickly,
but it mitigates the overfitting problem. I believe I can improve upon it with
other techniques.
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3. Adopting SwiGLU - various changes

I began switching my networks from regular Residual MLP to SwiGLU after learning about it in class. My initial implementation in Change 9 was horrendous because I'd failed to add the residual stream back to the backbone. In that case, the test accuracy fell below baseline. I fixed the issue in Change 10 and saw accuracy increase back above baseline to 53.27%.

3. Training dynamics.

Your ImageNet analysis from the section above: the figures and what they show about how your model trains. In the 5000-level section, this includes your ablation study.



Activation Variance Activation variance appears to be stable throughout the model. I see somewhat linear growth across blocks and as training continues, which is expected since we're adding variance to the backbone through the blocks. The good news is that there's no evidence of exploding or vanishing variance.

Branch-to-stream ratio In my 30-epoch diagnostic run, I see the branch-to-stream ratio diminish overall in later epochs. However, in mid and late epochs, it appears the ratio remains mostly flat throughout the blocks. Some of the later blocks (blocks 80 - 90) actually have a higher ratio than the middle blocks. Although the ratio was small, it was enough that the network continued to learn at depth 90. It's possible going even deeper could eke out a little bit of improvement.

RMS gradient I don't see any evidence of exploding or vanishing gradients in this network. There's a steady decrease over the depth in each of the sampled epochs, but it's in line with normal network behavior. It's

maintaining a healthy gradient throughout the depth, which tells me the residual connection and normalization is working.

Update-to-weight ratio I'm not sure what to make of the update-to-weight ratio graph for this model. It's difficult to read. I believe with the SWiGLU layers (up, down, gate) all being graphed together, it's difficult to read. There's wild oscillation between a ratio of 10^-2 and 10^-3. A quick Google search says this behavior could indicate that the learning rate is too aggressive. However, my AdamW optimizer is initialized with a learning rate of 0.001, which seems quite small already. I'd need to study this more in future work.

Saturation - there's practically no saturation in this network, which is a nice perk of using a SwiGLU network. Neurons are much less likely to die and can self-recover. I don't have a graph for ReLU for comparison, but in this network, dead or saturated neurons are not a problem.

Ablation Study My SwiGLU imagenet32 network has a depth of 90 blocks. The ablation study shows that many layers throughtout the depth are contributing to the accuracy of the model. There's a slight concentration of improvment within the first 15 blocks, but I see continued incremental improvements of 0.5% to 1.5% interspersed with less-frequent accuracy regressions throughout the depth of the network. This tells me that the network's depth is providing some value, and it's possible an even deeper network could further improve accuracy.

I ran the same abalation code on the cifar10 dataset, and it showed a different result. The bulk of improvement happened in the first 20 blocks with the later blocks contributing relatively little. But with the imagenet32 dataset, it appears that the deep blocks play a relatively more important role than they did in cifar10.

4. What did not work.

Describe ideas that failed and your best explanation of why.

Bigger not always better

I tried various combinations of enormous width and depth, but they had minimal impact on test accuracy. Often it just led to slower training.

Data Augmentation is an art

RandomAffine augmentation was a bust, at least with my params. I also tried various scaling and rotation transforms when running cifar10. I believe there's a lot of value in the built-in pytorch v2 transforms, but it was a struggle to unlock the best possible combination. I finally settled on AutoAugment with RandomHorizontalFlip for both imagenet and cifar.

In general, regular old bugs cost time.

As I'm learning Deep Learning, and no matter how long I've been coding, bugs still creep into my code. I had a couple of particularly annoying bugs that cost me a lot of time. First, I simply failed to add the residual value back to the backbone when I was first implementing SwiGLU blocks. That caused the variance stream to disappear and stopped learning. That was before I'd added my telemetry code, so it wasn't immediately obvious that the network wasn't learning. It cost me several hours to debug.

The second major bug caused me to turn in my analysis a day late. In my diagnostics notebook, I had failed to run the data augmentation step, which is where I handle the scaling and standardization of the inputs. So, during the ablation study, I was feeding the network very large values, which caused accuracy to be nearly zero and rendered the ablation steps meaningless. To have only found the issue after hours+ of training the network, it was a frustrating setback that delayed making progress on other network improvements.

5. Across the three tasks.

How and why do your final models differ between MNIST, CIFAR-10, and ImageNet 32×32? What does that tell you about the limits of MLPs on images?

Mnist is a simple enough task that it's hard to do incorrectly. I initially spent the first few changes of my project checking the baseline model and comparing it to a single-layer FFN as Dr. Roberts described in the notes. I confirmed that even a very simple FFN with just one hidden layer could exceed 97% accuracy. Once I was satisfied with that, I restored MNIST as a residual MLP. On the day we talked about GELU and SwiGLU in class, I decided to make all three of the tasks use SwiGLU since that seems to be a network commonly used in industry. I kept that for mnist.

For both MNIST and CIFAR10, overfitting is a problem. A careful application of data augmentation helped with that as I was able to use 30-deep SwiGLU blocks without harming testing too much. I kept MNIST and CIFAR10 as 30-deep networks in the final models.

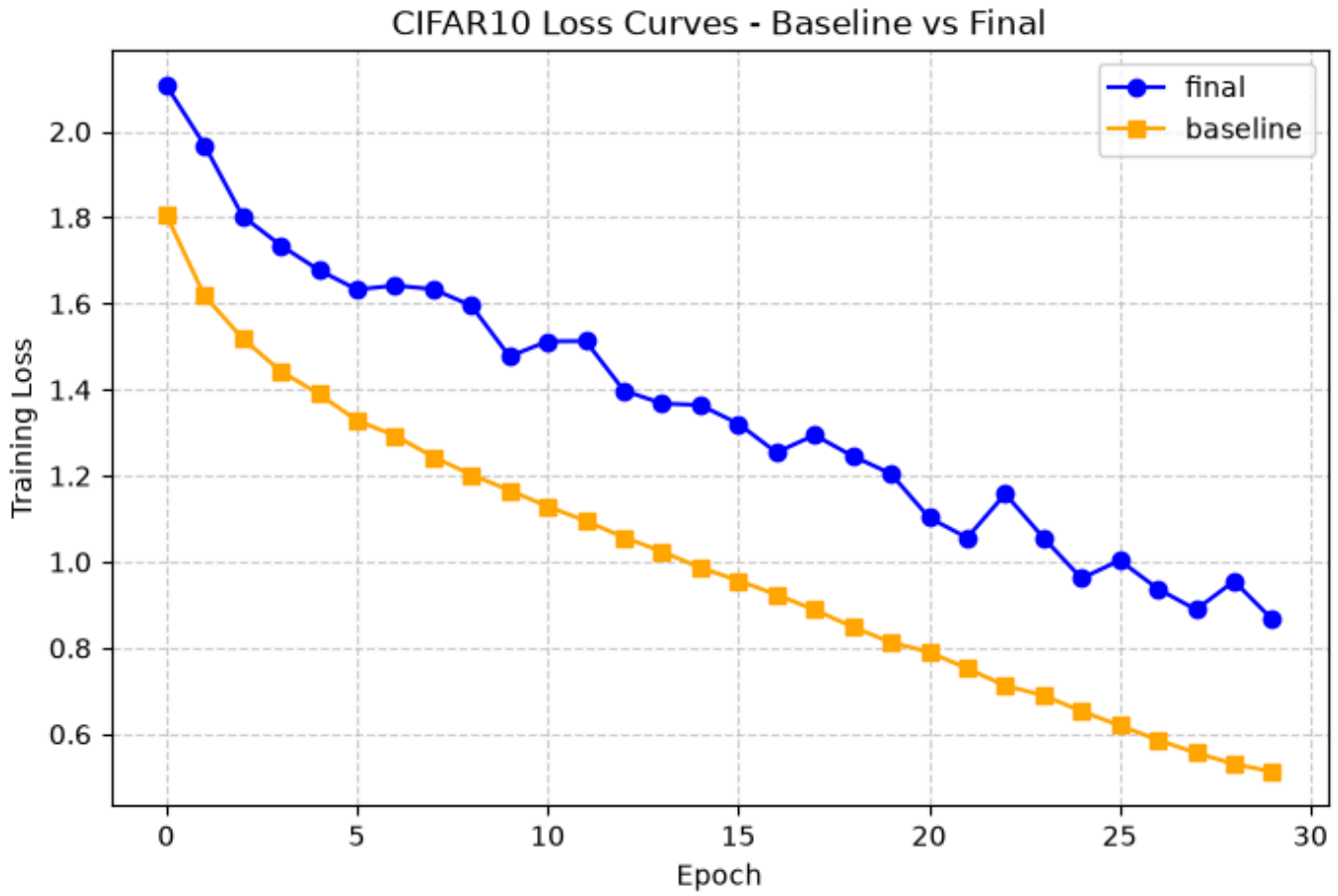
IMAGENET32 and CIFAR10 both used very similar data augmentation. Both had AutoAugment policies built-in to PyTorch, and I made use of that in addition to Horizontal Flip. For both networks, I kept width at 128 and hidden at 256 for the final models.

Even in the final models, CIFAR10 began exhibiting memorization behaviors around epoch 20 where training loss continued to drop quickly but test accuracy plateaued. Fortunately, I didn't see test accuracy deteriorate beyond that, so I kept training for 30 epochs.

The main difference in IMAGENET32 compared to CIFAR was the additional depth. In the final model, I pushed IMAGENET32 to a depth of 120 and still didn't see overfitting emerge.

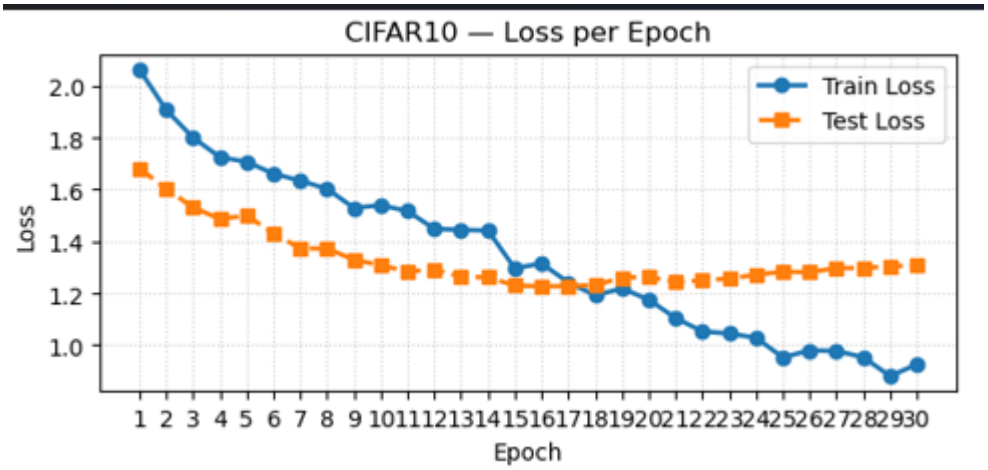
training curves for Cifar10 Baseline vs Cifar10 Final

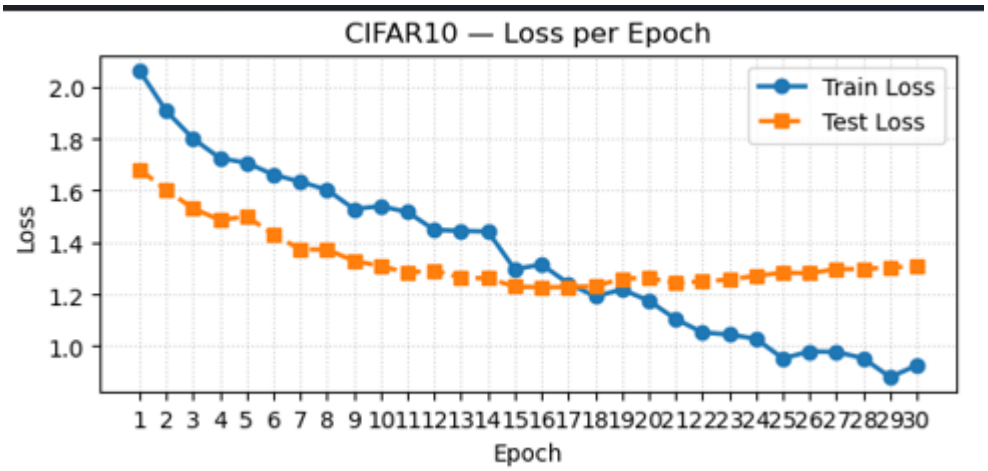
Compare CIFAR10 Loss Curves - Baseline vs. Final



Despite the baseline model appearing to converge more quickly, it's actually overfitting and does not perform as well as the final model.

CIFAR10 Final Loss and Accuracy Curves





6. Explanations of anything beyond the basics.

For each technique not covered in class, explain how it works and why you used it

I mainly stuck to techniques covered in class. I do wish I could've gone further but ran out of time.

In a future project, I'd like to experiment more with these other buiding blocks:

- GELU
- batch norm
- dropout
- label smoothing
- splitting image into patches